Restaurant Rating Prediction Using Machine Learning

Machine Learning Internship Project Report

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1. Introduction

Restaurant ratings influence customer decisions and business reputation. Predicting ratings using historical restaurant information can help restaurants understand customer satisfaction and improve their services. This project develops a regression model capable of estimating restaurant ratings from restaurant characteristics.

2. Objectives

- Predict restaurant aggregate ratings.
- Preprocess and clean the dataset.
- Perform exploratory data analysis.
- Compare multiple regression algorithms.
- Evaluate models using regression metrics.
- Identify the most influential features affecting ratings.

3. Dataset Description

Dataset Size: 9551 restaurants
Total Features: 21
Target Variable: Aggregate Rating

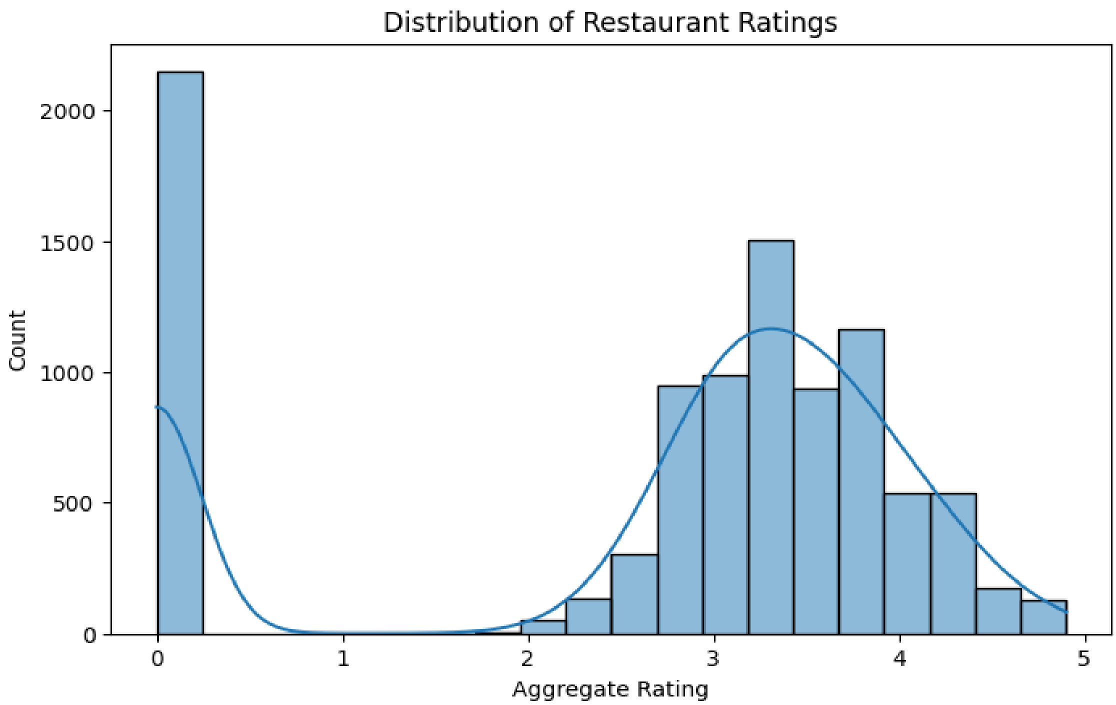
Important attributes include City, Cuisine, Average Cost for Two, Currency, Table Booking, Online Delivery, Price Range and Votes.

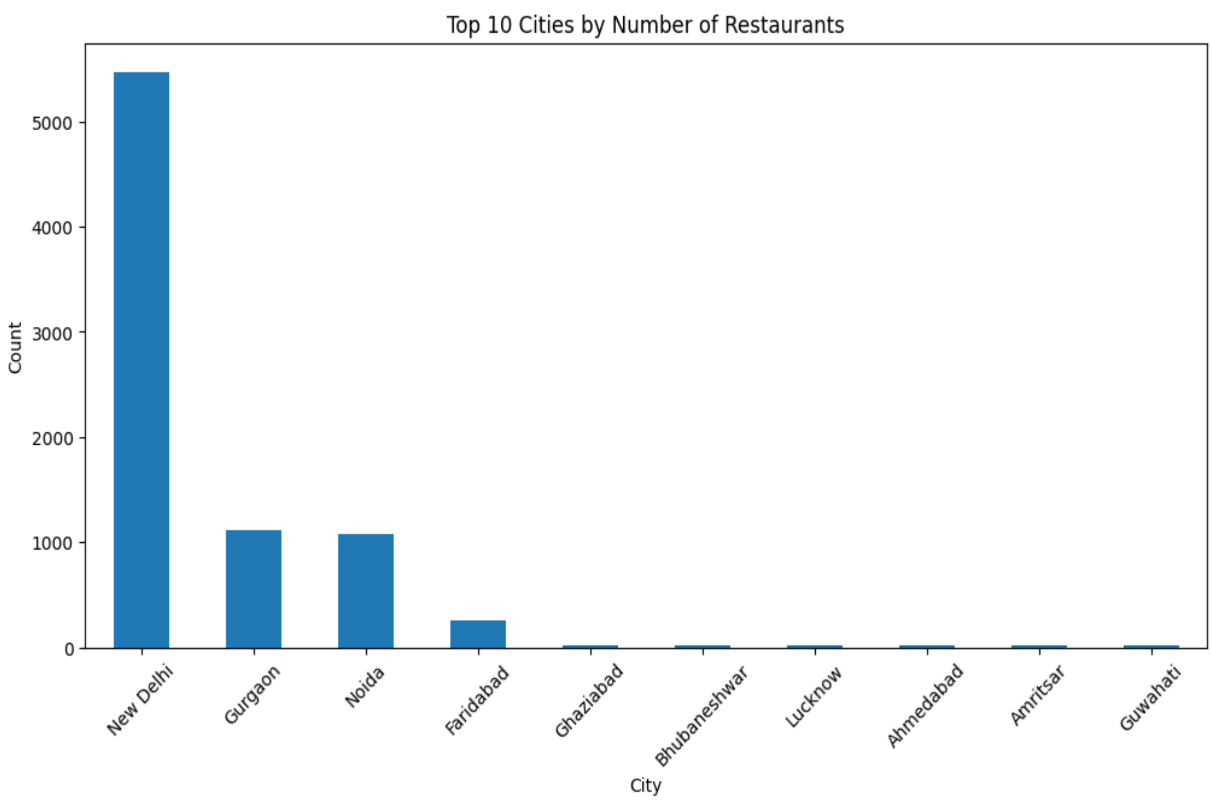
4. Data Preprocessing

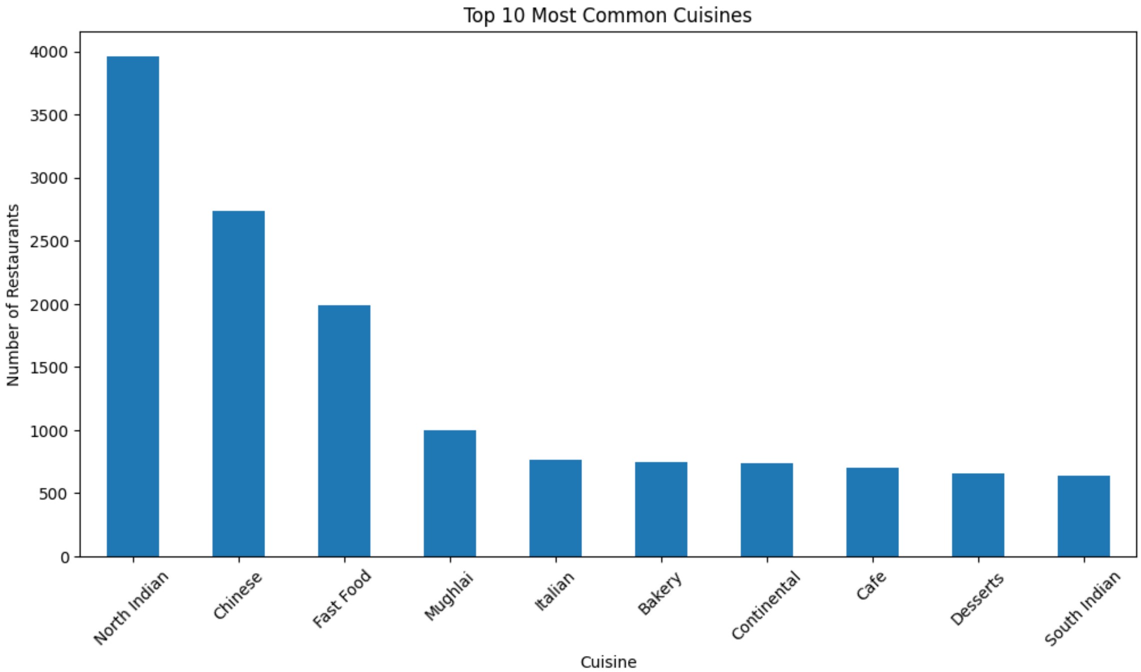
Missing values in the Cuisines column were handled by replacing them with 'Unknown'. Duplicate records were checked and none were found. Unnecessary columns such as Restaurant ID, Restaurant Name, Address, Rating Color and Rating Text were removed. Categorical features were encoded using LabelEncoder and the dataset was split into 80% training and 20% testing data.

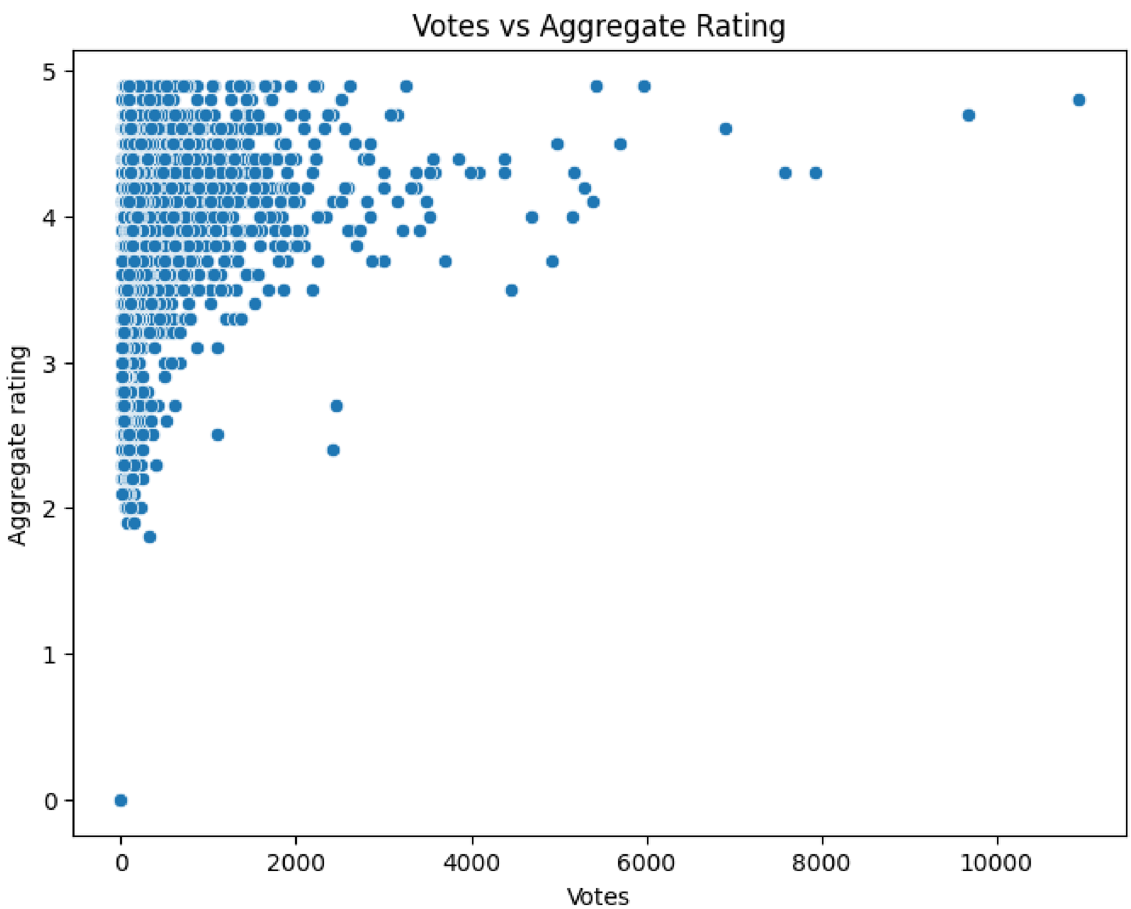
5. Exploratory Data Analysis

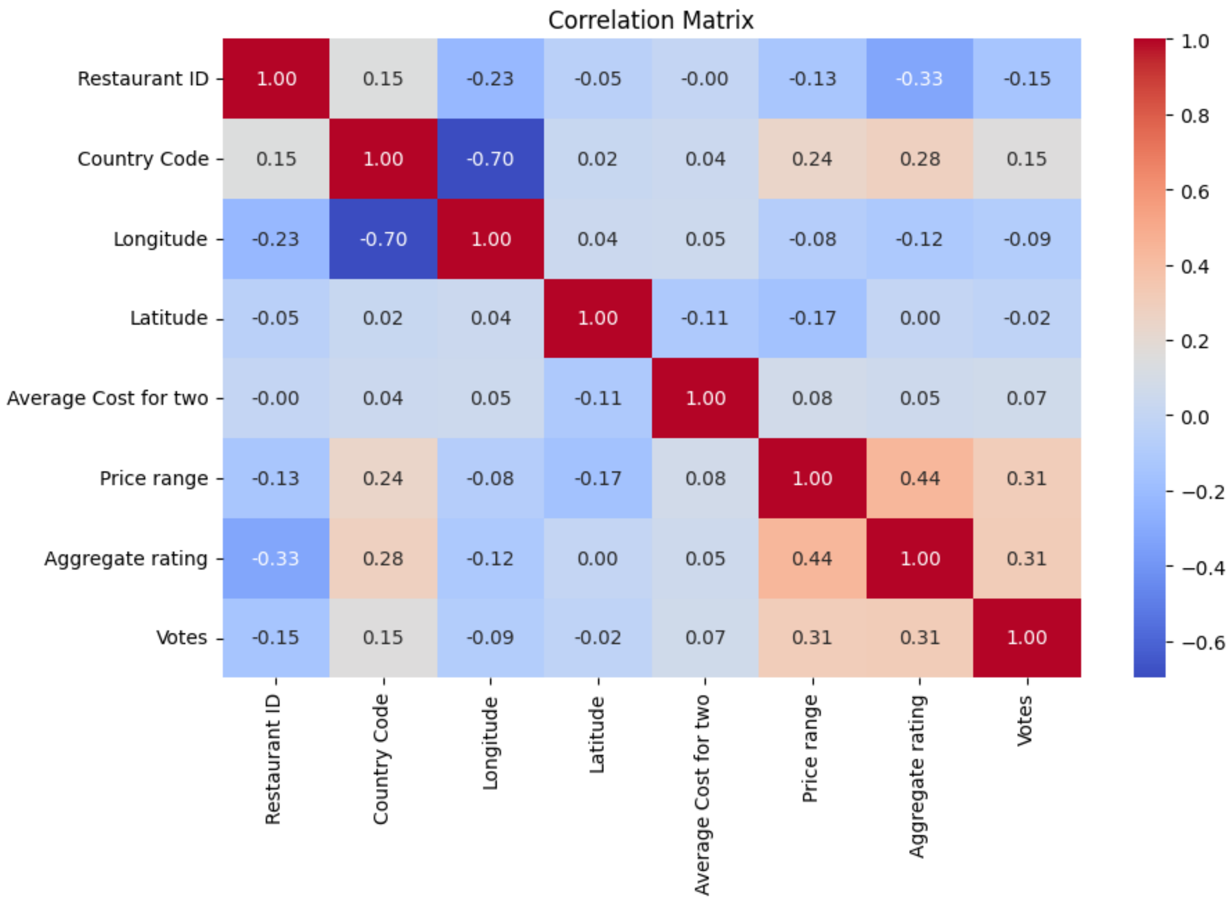
EDA was performed to understand data distribution and relationships among variables. The following visualizations should be inserted in this section:


Distribution of Restaurant Ratings
2000
1500
1000
500
0
Count
0
1
2
3
4
5
Aggregate Rating


Top 10 Cities by Number of Restaurants
5000
4000
3000
2000
1000
0
Count
New Delhi
Gurgaon
Noida
Faridabad
Ghaziabad
Bhubaneshwar
Lucknow
Ahmedabad
Amritsar
Guwahati
City


Top 10 Most Common Cuisines
Number of Restaurants
0
500
1000
1500
2000
2500
3000
3500
4000
North Indian
Chinese
Fast Food
Mughlai
Italian
Bakery
Continental
Cafe
Desserts
South Indian
Cuisine


Votes vs Aggregate Rating
Aggregate rating
0
1
2
3
4
5
0
2000
4000
6000
8000
10000
Votes



6. Machine Learning Models

Three regression algorithms were implemented:

1. Linear Regression
2. Decision Tree Regressor
3. Random Forest Regressor

Random Forest performed best because it captures nonlinear relationships and reduces overfitting through ensemble learning.

7. Model Evaluation

Performance Comparison

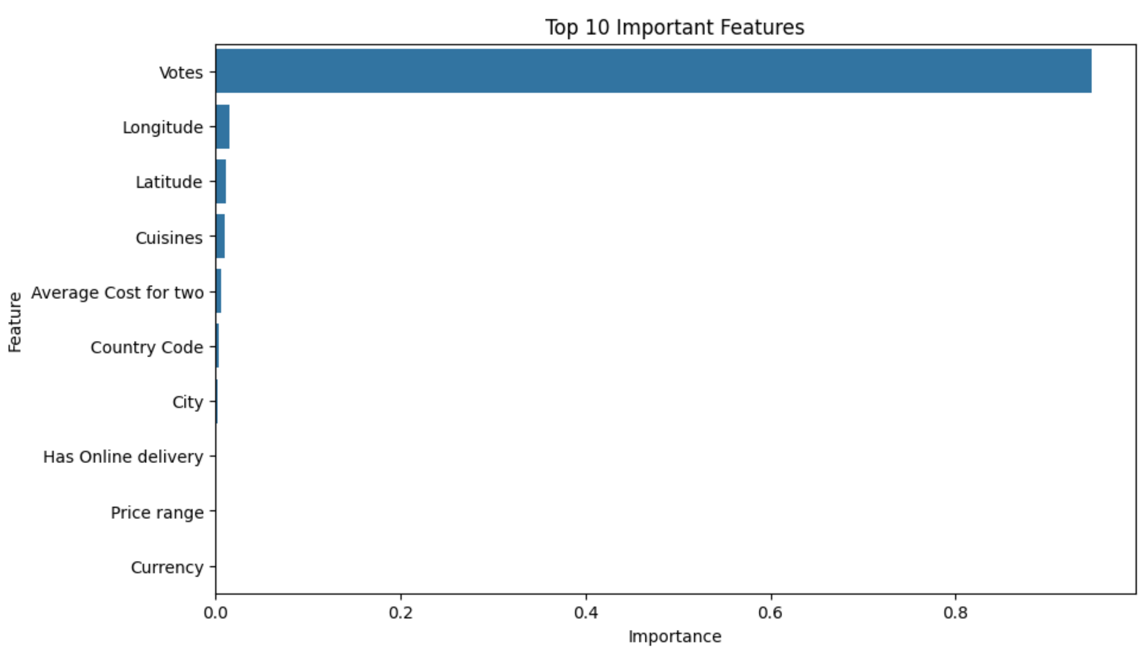
	Model	R2 Score	RMSE
2	Random Forest	0.961580	0.295715
1	Decision Tree	0.922681	0.419507
0	Linear Regression	0.303112	1.259441

The Random Forest model achieved the highest predictive accuracy and was selected as the final model.

8. Feature Importance

Top Important Features:

1. Votes
2. Longitude
3. Latitude
4. Cuisines
5. Average Cost for Two



	Feature	Importance
12	Votes	0.947476
2	Longitude	0.015274
3	Latitude	0.011814
4	Cuisines	0.010410
5	Average Cost for two	0.006682
0	Country Code	0.003242
1	City	0.001776
8	Has Online delivery	0.001139
11	Price range	0.000936
6	Currency	0.000662

Customer votes contributed the most towards predicting restaurant ratings, indicating that customer engagement strongly influences restaurant quality.

9. Conclusion

The project successfully developed a restaurant rating prediction model using machine learning. Random Forest Regressor outperformed the other algorithms and achieved excellent prediction performance. The project demonstrates a complete machine learning workflow from preprocessing to model evaluation.

10. Future Scope

Future improvements include hyperparameter tuning, One-Hot Encoding, Cross Validation, Gradient Boosting models, XGBoost, LightGBM and deployment as a production-ready web application.

References

1. Scikit-learn Documentation
2. Pandas Documentation
3. NumPy Documentation
4. Matplotlib Documentation
5. Seaborn Documentation